DreamLog

Compression Enables Generalization in Logic Programming

Alexander Towell

10.5281/zenodo.19490027 (preprint)

Southern Illinois University Edwardsville

April 2026

The idea

Problem. Knowledge bases grow but do not learn. You can add 300 facts about a family tree, and the system still cannot tell you whether a new person is someone's father. The facts accumulate, understanding does not.

Claim. Compression *is* learning (Solomonoff's idea). The shortest KB consistent with the data generalizes best.

System. DreamLog, a Prolog variant with wake-sleep cycles inspired by DreamCoder.

- **Wake:** answer queries against the KB, and if a predicate is undefined, optionally ask an LLM to propose rules for it.
- **Sleep:** run 8 compression operations (7 symbolic, 1 LLM-assisted), each verified against positive and negative test queries with atomic rollback.

Every compression is behaviorally equivalent to the original KB on the verification suite. Nothing gets "worse."

What “compression discovers concepts” looks like

From the crafting experiment, symbolic compression only (no LLM).
Input: seven ground facts, one per master artisan.

```
(master_artisan kael)    (master_artisan sera)
(master_artisan yara)    (master_artisan dax)
(master_artisan mira)    (master_artisan orin)
(master_artisan zara)
```

After one sleep cycle:

```
(master_artisan X) :- (artisan X),
                      (not (exception_master_artisan X)).
(exception_master_artisan thom).
(exception_master_artisan fen).
```

Seven facts become three clauses. The compression is accepted because one rule plus two exceptions is shorter than seven facts.
At test time, adding `(artisan venn)` makes `(master_artisan venn)` derivable. Nobody wrote that fact; the system classified a new entity using a concept it discovered from structure alone.

The result I care most about

The worked example above is one rule from the crafting experiment. Aggregate over 33 test checks (19 positive, 14 negative), 5 runs per condition:

Condition	Recall	Precision	Rules
No dream	0%	—	0
Raw LLM prompting	0%	—	0
Symbolic only	53%	59%	5
Full pipeline	$64 \pm 2\%$	64%	20

The raw LLM scores 0% because the terms are invented. Symbolic compression alone hits 53% without any LLM involvement, which means compression discovers real concepts from data structure, not from prior knowledge. The LLM adds another 11 points on top.

This is the cleanest evidence I have for the thesis.

The ask

The paper needs work before it's venue-ready: a head-to-head Popper or Metagol baseline, figures, and likely a larger-scale run.

What coauthoring might look like. Where I think you could contribute:

- Statistical rigor on the empirical claims (your ransomware work set a bar for how to report these)
- Venue strategy; a senior coauthor materially changes what's reachable
- Framing with respect to my dissertation: is this a chapter or a side project?
- Advocating for a Popper benchmark run, mostly student-time effort but needs to be scoped

What I'd do. Everything else: bugs, experiments, writing, figures, the ILP benchmark setup, revisions.